Where California's Farm Water Goes

A look at the Central Valley — and how much water better irrigation timing could save

Prepared 28 September 2026 · All figures are rounded estimates. Sources on the last page.

25 MAF	15 MAF	10 MAF	1 in 5
Put on Central Valley farms each year	What the crops actually drink	The gap between those two numbers	Gallons that never reach a crop

MAF = million acre-feet. One acre-foot is about 326,000 gallons — a football field under a foot of water.

1. The short version

California farms use about 80% of the water people use in this state. The Central Valley is the heart of it: roughly 6.5 million irrigated acres from Redding down to Bakersfield.

Farmers here put about **25 million acre-feet** of water on their fields every year. Their crops only drink about **15 million**. The rest soaks past the roots, dries into the air, or runs off the end of the row.

Some of that gap cannot be helped. Much of it comes down to **when you water and for how long** — and on about half of California's acres, that is still decided by the calendar or by habit.

2. Where the water goes

Here is what happens to every 100 gallons put on a field. The numbers change a lot depending on the irrigation system.

Where it ends up	Drip & micro (56% of CA acres)	Sprinkler (~14%)	Flood & furrow (~30%)	CA average
The crop drinks it	87 gal	78 gal	65 gal	**79 gal**
Soaks past the roots	6	7	22	**11**
Dries off the wet soil	5	3	5	**5**
Blows away or wets leaves	2	9	0	**2**
Runs off the field	0	3	8	**3**

Flood and furrow lose the most, and California still waters about 30% of its acres that way. Drip is much tighter, but not loss-free.

3. Which of those losses can be fixed by better timing?

Not all of them. It is worth being clear about which is which.

The loss	How big	Can timing fix it?	Why
Soaks past the roots	11 gal in 100	**Yes — mostly**	This is over-watering. It happens when a set runs longer than the soil can hold, or comes back before the soil has dried out.
Runs off the field	3 gal in 100	**Yes — partly**	Happens when water goes on faster than the ground can soak it up. Splitting a long set into shorter ones with rests fixes much of it.
Blows away / wets leaves	2 gal in 100	**Yes — partly**	Watering at night instead of midday roughly halves this loss on sprinklers (about 8% at night vs 15% in the day).
Dries off the wet soil	5 gal in 100	Barely	Depends mostly on how much ground gets wet, which is a hardware question, not a timing one.
The crop drinks it	79 gal in 100	No — and you would not want to	This is the whole point. Cutting here costs yield.

Added up, the losses that timing can reach come to roughly **10 to 16 gallons out of every 100** put on a California field.

4. What Farmify does about it

- **It works out what the crop actually needs today.** Not what it needed in July. Crop thirst in Fresno swings about eight-fold between January and July.
- **It waters when the soil is dry, not when the calendar says.** That might be every four days in summer and every two weeks in spring.
- **It never fills the soil past what it can hold.** In our test almond block, the plan sent **0.00 inches** past the roots across a whole season.
- **It puts every set at night.** All 25 irrigations in that same test block were scheduled between 6 pm and 8 am, which cuts the blow-away loss roughly in half.
- **It splits sets that go on too fast**, so water soaks in instead of running off.
- **It tells you in plain words**, in the language spoken on the farm, on one page you can print and hand to whoever opens the valve.

5. How much would that save?

We ran Farmify's own engine on six typical Central Valley crops, then re-ran the same crops, soils and equipment under two common ways of scheduling. Only the timing changed.

How the watering was scheduled	Water used	Compared to Farmify
Farmify — waters to what the crop needs	236 in	—
Adjusts monthly, plus a bit extra to be safe	278 in	used **15% more**
One setting, left alone all season	409 in	used **42% more**

No crop went thirsty in any of the three. The water was saved by not wasting it, not by under-watering.

Which number is realistic?

Most Central Valley farmers are somewhere in the middle. Real almond orchards here use about 4.0–4.5 acre-feet per acre, and Farmify's plan came out at 4.22 — so average practice is a lot closer to the second row than the third.

A realistic saving is 10–20% of the water a farm puts on. That matches outside research on ET-based watering, which usually finds 10–30%.

6. What that is worth

For one 100-acre almond farm (which uses about 422 acre-feet a year):

If it cuts	Water saved	That is	Pumping cost saved each year
10%	47 acre-feet	15 million gallons	$7,000 – $14,000
15%	74 acre-feet	24 million gallons	$11,000 – $22,000
20%	106 acre-feet	34 million gallons	$16,000 – $32,000

Cost assumes $150–300 per acre-foot to pump and deliver. In a drought year water can cost several times that.

For the whole Central Valley, if the half of acres not using crop water data started:

If those farms cut	Water saved each year	For comparison
10%	about 1.2 MAF	roughly two-thirds of the valley's groundwater overdraft
20%	about 2.5 MAF	more than the entire overdraft

The valley pumps about 1.8–2.0 MAF more groundwater than gets replaced each year, and state law (SGMA) requires that closed by 2040. Some districts already cap growers at 1.6 acre-feet per acre — against the 3–4 an almond orchard wants.

One honest catch

Water that soaks past the roots here mostly drips back into the same groundwater the farmer pumps from. It is wasted *effort* — pumped, pressurised and paid for — but not all lost *water*. Counting only what truly disappears, the real valley saving is more like **0.4 to 1.2 MAF a year**. The money, fuel and fertiliser saved are real either way.

7. The point

Farmify does not make new water. It takes a decision that half the valley currently makes by guesswork, does the arithmetic properly, and hands back a plan anyone can follow.

Where these numbers come from

- **Central Valley water balance** (25 MAF applied, 15 MAF crop use, 6.8M irrigated acres) — US Geological Survey, *Irrigation Water Use in the Central Valley of California*.
- **California water split and groundwater overdraft** — Public Policy Institute of California: *Water Use in California's Agriculture*, *Groundwater in California*, *A Reality Check on Groundwater Overdraft in the San Joaquin Valley*.
- **Irrigation system mix** (56% of CA acres on drip or micro, ~30% still flood) — USDA Economic Research Service irrigation data, 2023.
- **System efficiency** (furrow ~65%, sprinkler ~85%, buried drip ~95%) — UC Davis Water Management and USDA NRCS irrigation guides.
- **Blow-away and evaporation, day vs night** (15.4% day, 8.5% night on solid-set sprinklers) — published sprinkler wind-drift and evaporation loss measurements.
- **How farmers actually schedule** — USDA irrigation survey; California almond grower practice survey (only 38–55% use crop water data or soil moisture).
- **Sacramento Valley** (~2M irrigated acres, rice about a quarter of them) — Sacramento River Watershed Program; PPIC.
- **The crop water maths inside Farmify** — FAO Irrigation and Drainage Paper 56 (Allen, Pereira, Raes & Smith, 1998), calibrated against California's CIMIS weather station network. Weather history from Weather Underground.

The comparison in section 5 was produced by running Farmify's own engine, not taken from someone else's study. The 100-gallon breakdown in section 2 is built from published efficiency figures for each system type, weighted by how much California acreage uses each one. Treat everything here as a good estimate, not a measurement of any one field.